

HaptingleXR: Haptics-based Mixed Reality System Designed to Reduce Anxiety and Pain during Intravenous Injection Procedures

Youngbin Kong*

Interdisciplinary Program in Medical Informatics
Seoul National University College of Medicine
Seoul, Republic of Korea
Department of Transdisciplinary Medicine
Seoul National University Hospital
Seoul, Republic of Korea
ybkong@snu.ac.kr

Kyu Eun Lee

Department of Surgery
Seoul National University Hospital
Seoul, Republic of Korea
Department of Surgery
Seoul National University College of Medicine
Seoul, Republic of Korea
Medical Big Data Research Center
Seoul National University College of Medicine
Seoul, Republic of Korea
kyueunlee@snu.ac.kr

Yeonsu Seo*

Department of Transdisciplinary Medicine
Seoul National University Hospital
Seoul, Republic of Korea
ynsoo.seo@gmail.com

Hyoun-Joong Kong[†]

Department of Transdisciplinary Medicine
Seoul National University Hospital
Seoul, Republic of Korea
Institute of Convergence Medicine with Innovative
Technology
Seoul National University Hospital
Seoul, Republic of Korea
Department of Medicine
Seoul National University College of Medicine
Seoul, Republic of Korea
gongcop7@snu.ac.kr

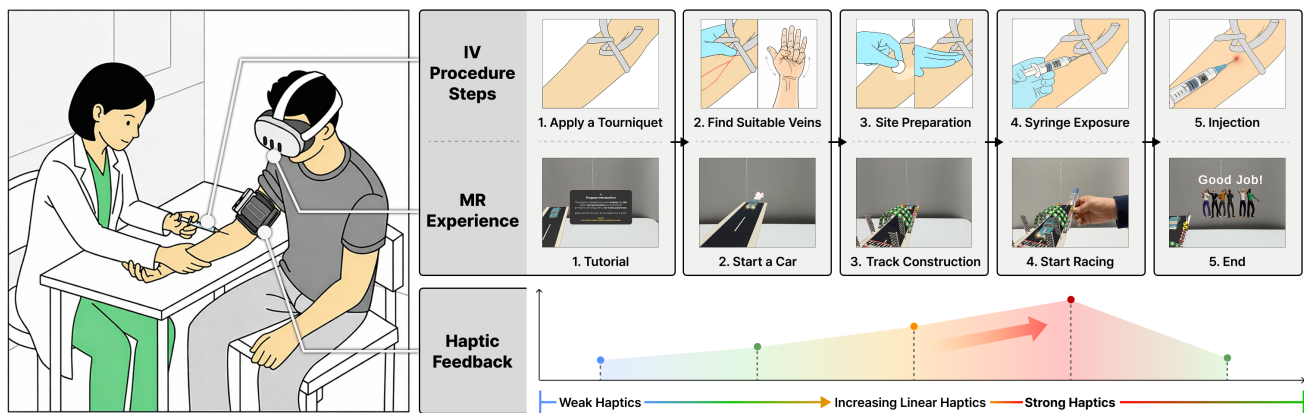


Figure 1: Overview of HaptingleXR, a haptics-based mixed reality system that aligns visual and haptic interaction with the intravenous injection workflow to reframe the procedural experience.

*Both authors contributed equally to this research.

[†]Corresponding author



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Abstract

Intravenous (IV) injection is a routine medical procedure often associated with anxiety and pain, affecting procedural efficiency and patient experiences. To address this, previous XR-based approaches have relied primarily on visual distraction, and physical

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interventions have offered tactile stimulation for pain mitigation. However, neither integration sufficiently integrates with the procedural flow nor accounts for the embodied nature of the injection. We introduce HaptingleXR, a haptics-based mixed reality system designed to address injection-related anxiety and pain by reframing the injection experience. HaptingleXR positions the procedure as a goal-structured activity that aligns with the injection's workflow. The experience takes the form of a racing scenario in which patient movements are reflected in a mixed reality environment. Synchronized haptic feedback is provided during key procedural moments along with visual interaction. We describe the system design and implementation and outline this mixed reality system as an experience-centered approach that may support patient participation during medical treatments.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality; Haptic devices.**

Keywords

Mixed Reality, Haptic Feedback, Intravenous Injection Procedures, Anxiety Reduction, Pain Reduction

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1 Introduction

Intravenous (IV) injection is widely performed as an essential medical procedure for diagnosis, treatment, and nutritional support in clinical settings [2]. Globally, more than one billion peripheral IV injections are administered annually to hospitalized patients [15], with up to 80% of inpatients receiving IV injections during their hospital stay [14]. This procedure is indispensable for patients requiring emergency care or perioperative management [1].

Patients experience significant anxiety and pain during IV injection procedures [8, 21]. Recent studies indicate that 73.3% of patients experience anxiety during IV injection, and 95% report pain [27]. Notably, the psychological anxiety preceding the actual injection often causes more stress than the physical pain of the needle insertion itself [10, 20]. Excessive anxiety can cause vasoconstriction, leading to lower success rates in IV injection procedures [25]. For patients with difficult IV access, failed attempts can delay treatment and increase operational burden, with broader implications for healthcare quality and cost [16, 20]. In particular, for patients with chronic conditions or those requiring regular treatment, repeated negative experiences can lower treatment compliance, potentially affecting long-term treatment results [17].

Extended reality (XR) technologies have been increasingly explored as interventions to reduce anxiety during medical procedures. Prior studies have primarily focused on immersive visual environments using virtual reality (VR) to distract patients from

anxiety-inducing stimuli. Randomized controlled trials have reported that VR can reduce perceived anxiety and pain during intravenous procedures [11, 19]. However, many XR-based interventions rely predominantly on perceptual diversion and position patients as passive recipients of immersive content, with limited integration into the body and flow of clinical practice.

In parallel, vibration-based interventions grounded in the Gate Control Theory of pain [18] have demonstrated that non-noxious tactile stimulation can attenuate nociceptive transmission [13]. Devices such as Buzzy (MMJ Labs, Atlanta, Georgia) have been shown to reduce perceived pain intensity during needle-related procedures [3]. While these approaches provide evidence for sensory modulation effects, they primarily target pain signaling, leaving the broader experiential structure comparatively underexamined.

Prior research on mental health gamification emphasizes the careful integration of therapeutic and game elements [22]. Studies investigating XR for anxiety reduction further suggest that preserving bodily awareness and perceived agency, rather than relying solely on visual immersion or distraction, may better support meaningful engagement [4, 5]. Additionally, interactive VR designed to reduce anxiety during MRI procedures demonstrates that active participation and real-time feedback can enhance procedural preparedness compared to passive immersion [26]. These studies collectively highlight the need for interventions that shape the experience of clinical procedures beyond simple distraction.

Building on these insights, we present HaptingleXR, a haptics-based mixed reality system designed to reduce anxiety and pain during IV injection. HaptingleXR operationalizes a behavioral reframing framework by integrating synchronized multisensory feedback into the clinical workflow, restructuring the procedural experience within a goal-directed, game-like context. While not claiming immediate clinical efficacy, we introduce HaptingleXR alongside a study design to investigate how this approach may influence anxiety and pain during IV injection and to explore the broader potential of mixed reality as a procedure-structuring framework in clinical settings.

2 Design of HaptingleXR

HaptingleXR implements this reframing framework through three coordinated design strategies embedded within the clinical workflow: goal-structured progression, embodied hand–arm coupling, and temporally aligned multisensory feedback.

2.1 Goal-Structured Racing Interaction

Racing serves as the central interaction metaphor because of its psychological alignment with goal-oriented motivation [9, 24]. Racing emphasizes reaching a clear endpoint, monitoring progress, and achieving success. The visible finish line reframes the injection, shifting the perception from an uncontrollable event to a bounded and progressive one.

By introducing a goal-directed structure, the system shifts the patient's focus from anticipating pain to achieving their goal. This temporal structuring may reduce uncertainty and waiting-related stress, both of which are known contributors to procedural anxiety.

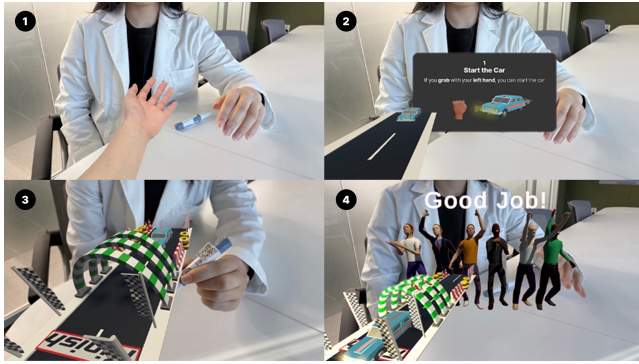


Figure 2: Screenshots of the HaptingleXR prototype in a simulated IV injection setting. Virtual racing elements are rendered on the arm and progress in alignment with the procedural workflow.

2.2 Embodied Hand-Arm Coupling

A core design decision in HaptingleXR was the direct mapping between the patient’s physical body and the mixed reality racing environment. Clinically required actions, such as clenching the fist, are mapped to starting the car engine, whereas the patient’s arm itself becomes the racing track. This mapping is intended to reinforce the embodied agency rather than function as a superficial gameplay mechanic.

Unlike virtual reality, which often extracts users from their physical surroundings, mixed reality allows patients to remain perceptually grounded in the real clinical environment while directly observing their bodies interacting with augmented elements [5]. This direct perception of one’s physical body maintains contextual awareness and supports a stronger sense of agency, factors that have been associated with reduced anxiety responses in medical settings.

By positioning the patient’s arm as the structural anchor of the experience, HaptingleXR reframes the injection not as an event imposed on the body but as an activity enacted through it. This embodied alignment supports perceived control in medical contexts, where loss of control is a primary source of distress.

2.3 Sensory Alignment With Clinical Workflow

The multisensory system integrates visual immersion with temporally synchronized haptic feedback, addressing the limitations of XR-based medical interventions that rely primarily on visual distraction. Prior research suggests that suppressing visual input alone may heighten sensitivity to other sensory channels, such as ambient sounds or needle-related tactile sensations, potentially increasing anxiety [4]. HaptingleXR coordinates visual and tactile modalities that evolve together across the injection timeline.

The mixed reality experience follows the temporal structure of intravenous injection without disrupting standard clinical practices. Clinically required actions are mapped to the corresponding events within the racing scenario. Tourniquet application and fist pumping for vein detection constitute the preparatory phase, accompanied by a visual engine start effect. Subsequently, the track is progressively

constructed in synchrony with site preparation actions, such as alcohol swabbing. When the syringe enters the patient’s field of vision, a finish line appears, and the race begins. As the needle insertion proceeds, the race concludes with completion feedback.

Haptic feedback is structured into three modulated intensity stages corresponding to each injection phase. During a brief tutorial phase, patients are exposed to all intensity levels to establish familiarity with the device. Low-intensity vibrations are triggered by each fist pump during the preparatory phase, producing vibration bursts that reinforce the engine-start visual feedback. Moderate feedback occurs during track construction and is delivered in a linear spatial pattern along the arm, synchronized with in-game animations in which the road elements are sequentially placed. This spatial progression parallels the linear motion of alcohol swabbing before injection. High-intensity stimulation is delivered from the moment the syringe enters the patient’s field of vision until immediately before needle insertion, providing strong non-noxious tactile input intended to prime somatosensory pathways ahead of needle contact.

By aligning visual progression, spatialized haptic feedback, and injection timing, the system distributes the attention across the coordinated sensory streams. This multisensory structuring aims to support engagement and may contribute to reducing pain and anxiety while reinforcing the perceived control.

3 System Implementation

HaptingleXR was implemented in Unity and deployed on a Meta Quest 3 using the Meta XR SDK for optical hand tracking. The mixed reality application ran directly on the headset in standalone mode.

The system consisted of a mixed reality scene rendered on a head-mounted display, a wearable haptic device attached to the upper arm, and a state controller implemented within Unity. The state controller monitored the procedural progress and triggered the corresponding visual and haptic events based on the detected inputs.

Hand tracking was used to detect the interaction gestures. A grabbing gesture initiated the racing sequence within the scene and triggered the corresponding state transitions.

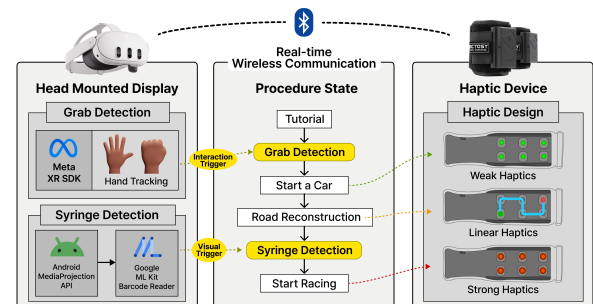


Figure 3: System architecture of HaptingleXR. Grab detection (hand tracking) and syringe detection (marker-based vision) trigger state transitions. The updated state coordinates visual progression and stage-specific haptic feedback.

Syringe exposure was detected using a marker-based visual-sensing system. The headset display stream was accessed through the Android MediaProjection API, and markers attached to the syringe were detected using the Google ML Kit Barcode Reader. When an exposure event was detected, the system state was updated and the corresponding visual and haptic outputs were activated.

Haptic feedback was delivered using the bHaptics Tactosy for Arms (bHaptics Inc., Seoul, Republic of Korea), which was worn on the upper arm and equipped with six vibration motors arranged in a 3×2 grid pattern. Communication between Unity and the device was established using the bHaptics SDK via Bluetooth. Three predefined vibration intensity levels were configured and activated based on the procedural stage. All interactions and feedback were coordinated within Unity to maintain consistent timing across the modalities.

4 User Study Plan

Study Design. We plan to conduct a between-subject randomized controlled study to evaluate the effectiveness and acceptability of HaptingleXR compared with standard care during intravenous injection. This study targets patients with needle-related anxiety, a clinically relevant population for whom anxiety-reduction interventions are particularly needed.

Research Questions. This study addresses the following research questions:

- **RQ1:** Does HaptingleXR reduce self-reported anxiety compared to standard care?
- **RQ2:** Does HaptingleXR reduce self-reported pain during intravenous injection?
- **RQ3:** How do patients perceive the usability and acceptability of HaptingleXR in a medical context?

Participants. We will recruit 40 adult patients (20 per condition) scheduled for peripheral IV injections. Eligible participants will be aged 19–65 years with a score of 35 or above on the Multidimensional Fear-of-Injection Scale (MFIS) [12], indicating the presence of injection fear. Participants must be able to wear a head-mounted display and complete the study procedures. The exclusion criteria will include severe motion sickness, seizure disorders, significant sensory or cognitive impairments, and emergency or critical care status.

Conditions. Participants will be randomly assigned to either the HaptingleXR intervention or a standard care control condition. Owing to the nature of the intervention, blinding of the participants and clinical staff is not feasible.

Measures. The primary outcomes will include state anxiety, measured using the short-form State-Trait Anxiety Inventory-State (STAI-S) [23] and pain intensity, measured using a 100-mm Visual Analog Scale (VAS) [7]. Secondary measures will include the System Usability Scale (SUS) [6], perceived presence, cybersickness (single-item scales), and brief, semi-structured interviews.

Procedure. After screening and informed consent, the participants will complete baseline assessments prior to injection. The intervention group will experience HaptingleXR throughout the procedure, whereas the control group will receive standard care. Participants in the intervention group may request the removal of

the HMD at any point, after which standard care will continue without interruption. Post-procedure assessments will be performed immediately after the injection.

Data Analysis. Quantitative data will be analyzed using between-group comparisons, with the baseline MFIS score as a covariate. Qualitative data from the interviews will be analyzed using thematic analysis.

5 Limitations and Future Work

The proposed system was implemented as a functional prototype primarily intended for proof-of-concept validation. Procedural events, such as syringe exposure, were detected using marker-based sensing, which is effective at the prototyping stage but does not fully reflect the diversity of syringe types or the complex visual conditions encountered in real-world clinical settings. In addition, the scope of this study emphasizes the presentation of the system design, interaction concept, and evaluation framework rather than reporting extensive findings from the user studies.

To address these limitations, future studies should focus on the following improvements:

- **Object Detection-based Procedural Sensing:** Future work will replace the marker-based sensing mechanism with an object detection-based approach capable of recognizing real syringes in clinical settings.
- **User Study and Comprehensive Evaluation:** Building on the proposed experimental design, future studies will conduct user experiments to quantitatively and qualitatively assess the effects of HaptingleXR.

By addressing these directions, the system is expected to evolve into a more reliable and effective tool for medical environments, with the potential to improve patient experience and support more efficient clinical procedures.

6 Conclusion

This study presents HaptingleXR, a haptics-based mixed reality system designed to address anxiety and pain during intravenous injection through experiential reframing and multisensory interactions. Rather than relying on visual distraction alone, the system integrates embodied interaction and synchronized tactile feedback within the clinical workflow.

We describe the design rationale underlying the use of a racing metaphor, hand-arm mapping, and multisensory integration, as well as the system architecture that enables real-time coordination between virtual interaction and physical procedures. A randomized controlled study targeting patients with needle-related anxiety is planned to evaluate the clinical and experiential outcomes.

By reframing IV injection from a passive and anxiety-associated event into a goal-structured and participatory experience, HaptingleXR illustrates how mixed reality can serve as a procedure-structuring medium in medical contexts. This study contributes to human-computer interaction research by demonstrating how procedural reframing and multisensory integration can address emotional and sensory challenges in healthcare and suggests directions for future human-centered interventions that support patient engagement and clinical practice.

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